Bowen Yang

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Research Interests

Robot learning for manipulation: vision–language–action models, demonstration collection, and the sensing, calibration, and control layers beneath learned policies.

Education

Shanghai Jiao Tong University Aug 2023 –
B.A. in French; B.Eng. in Information Engineering (minor degree) Jun 2027 (expected)
SJTU Paris Elite Institute of Technology. Advisor: **Prof. Junchi Yan**.
Average score: **91.69**, rank **2/77**
Selected coursework (score/100): Mathematical Foundation for Artificial Intelligence (99), Advanced Linear Algebra (97), Advanced Electronics (97), Probability and Statistics (96), Data Structure (96), Computer Architecture (96), Optimisation (94).

Publications

GuidedVLA: Specifying Task-Relevant Factors via Plug-and-Play Action Attention Specialization 2026
Xiaosong Jia*, **Bowen Yang***, Zuhao Ge*, Xian Nie*, Yuchen Zhou*, Cunxin Fan*, Yufeng Li, Yilin Chai, Chao Jing, Zijian Liang, Qingwen Bu, Haidong Cao, Chao Wu, Qifeng Li, Zhenjie Yang, Chenhe Zhang, Hongyang Li, Zuxuan Wu†, Junchi Yan†, Yu-Gang Jiang†
Robotics: Science and Systems (RSS)

ELAN4D: Embodiment-Centric 4D Supervision for Vision-Language-Action Models via Plug-and-Play Adaptation 2026
Zeyuan He*, **Bowen Yang***, Zhirui Fang*, Keru Zhou*, Lei Jiang, Jingjing Qian, Fan Mo, Junchi Yan, Philip Torr, Xiu Li, Li Jiang, Jialin Yu†
Conference on Robot Learning (CoRL)

FOMI: A Force-Feedback Manipulation Interface for Teaching Contact-Rich Manipulation 2026
Yishu Shen, Yiming Luo, Ruilin Wang, **Bowen Yang**, Shuai Yang, Fangyi Xu, Yujun Shen, Fu Zhang, Boyu Zhou, Yinghao Xu†
arXiv preprint arXiv:2610.19839
* Equal Contribution; † Corresponding Author

Research Experience

RobbyAnt (Ant Group), Algorithm Engineer Intern Jun 2026 – Present
Contact-rich demonstration collection and policy learning (co-author, FOMI, arXiv)

- Built **80 Hz synchronized** multi-sensor acquisition with a **data-check tool**, and extended **tracker-to-robot calibration** (forward kinematics, nonlinear least squares).
- Collected **robot-free and teleop demonstrations** and built the **PICO 4 Ultra VR baseline**.
- Implemented **relative-pose actions** and **real-time chunking**. Co-trained **$\pi_{0.5}$ policies**.
- Deployed and evaluated policies on **Franka and Flexiv**: **78.4%** average success on four tasks (100 demonstrations each), and **68.2%** after transfer to a dual-arm Flexiv.

RethinkLab, Shanghai Jiao Tong University, Research Assistant Sep 2025 – May 2026
Advisors: Prof. Junchi Yan and Prof. Xiaosong Jia.

GuidedVLA, **co-first author** Sep 2025 – Jan 2026
Task-relevant representation learning for VLA policies, RSS 2026 (project page)

- Led **codebase implementation and simulation experiments** for explicit supervision of task-relevant factors in a vision–language–action policy.

- Implemented **specialized action-attention heads** for **object grounding**, **temporal skill recognition**, and **depth-based geometric perception** in the π_0 action decoder.
- Integrated a **ControlNet-style residual adapter** for plug-and-play factor supervision.
- Evaluated against π_0: average success rose from **68.2% to 75.4%** on LIBERO-Plus and from **77.4% to 90.6%** on RoboTwin 2.0.

ELAN4D, **co-first author** Mar 2026 – May 2026
4D supervision for VLAs, CoRL 2026, with **Torr Vision Group, University of Oxford**

- Proposed **URDF-based forward kinematics** to turn proprioception into future **robot-joint and end-effector keypoint tracks** as low-cost **4D supervision**.
- Proposed a **ControlNet-style residual branch** that **isolates auxiliary gradients** from the pretrained VLM and keeps the base policy's inference interface unchanged.
- Co-designed **ablations** on 4D supervision, prediction location, and **robot-keypoint versus whole-scene tracking** to isolate the effect of each component.
- LIBERO-Plus success rose from **53.6% to 67.6%** (π_0) and **73.6% to 78.2%** ($\pi_{0.5}$).

Selected Projects

RoboMaster University Championship, Team Jiao Loong Sep 2023 – Aug 2025
Vice-Captain & Algorithm Lead (Aug 2024 – Aug 2025)

- Led a **15-member algorithm team** that became **national champions in 2024 and 2025**. Also recruited and trained **60+ members** and ran a **32-team on-campus competition**.
- Developed **FAST-LIO2 LiDAR–inertial SLAM** with **GICP map registration** and **PTP-synchronized LiDAR timing** for localization during fast robot motion.
- Built a motion planner combining **kinodynamic A*** and **model predictive control (MPC)** for navigation and trajectory tracking on embedded hardware.
- Deployed **TensorRT YOLO** on **Jetson Orin**, with **EKF tracking** of occluded robots.

Honors & Awards

National Scholarship 2024, 2025
Ministry of Education of China

National Champion, RoboMaster University Championship 2024, 2025
Team Jiao Loong, Shanghai Jiao Tong University

RM Award (Algorithm), RoboMaster University Championship 2025
Individual technical honor; one of three algorithm recipients in 2025

Finalist, Mathematical Contest in Modeling (MCM) 2025

Bronze Medal, National Olympiad in Informatics (NOI) Winter Camp 2021
China Computer Federation (CCF)

Platinum Division, USACO 2020
USA Computing Olympiad (USACO)

Technical Skills

Programming: Python, C, C++
Machine Learning & Vision: PyTorch, OpenCV, TensorRT
Robotics & Development: ROS/ROS 2, Docker, Git, LaTeX
Robots & Hardware: Franka, Flexiv, Jetson Orin, LiDAR, PICO 4 Ultra
Simulation & Benchmarks: LIBERO, LIBERO-Plus, RoboTwin 2.0

Languages

Chinese: Native.
English: TOEFL iBT **5.5/6** (ETS comparable total: **110/120**), 8 Mar 2026.
French: DELF B2 (CEFR B2), diploma issued 26 Apr 2026.